Air University
Mid-Semester Examinations: FALL 2026

Student ID:

Subject: Data Structures
Class: BSCYS
Section(s): A, B,
Course Code: CS-216
Course Lead Sign:

Time Allowed: 120 Minutes
Max Marks: 50
FM's Name: Salma Bukhari
FM's Signature:
Chair Dept. Sign:

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.

Question 1	CLO 1	GA 2	Domain level Understanding	Marks = 15

a)

(i) Differentiate between an Abstract Data Type (ADT) and a data structure with a suitable example for each (2 Marks)

(ii) Define the Stack ADT by listing its operations with a one-line description of each, and name two different data structures that can be used to implement it. (3 Marks)

b) Write an algorithm rem_node(location) to delete the node at a given position from a singly linked list. Your algorithm must handle an empty list, an invalid location, deletion at the head, and deletion in the middle or at the tail. (5 Marks)

c) Briefly define Big-Oh (O), Big-Omega (Ω) and Big-Theta (Θ) notations (2 Marks). Then calculate the time complexity of the following pseudocode by showing the working for each block, and express the result in Big-Oh notation: (5 Marks)

```
ALGORITHM compute (n)
  count = 0
  for i = 1 to n
    for j = 1 to n
      count = count + 1
  for k = 1 to n
    count = count + k
  return count
```

Question 2	CLO 2	GA 3	Domain level Applying	Marks = 15

a) Write an algorithm that uses a stack to check whether an expression containing the brackets ( ), { } and [ ] is balanced. Dry run your algorithm on the expression:

{ [(a + b) * c] - d) }. (6 Marks)

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b) Write an algorithm to insert a new node at the end of a circular singly linked list. Handle the case when the list is empty. (4 Marks)

c) Write algorithms for the enqueue and dequeue operations of a queue implemented using a singly linked list (with front and rear pointers). Handle the underflow condition and the case when the queue becomes empty after a dequeue. (5 Marks)

Question 3	CLO 3	GA 3	Domain level Applying	Marks = 20

a) The Air University library stores book IDs in a Binary Search Tree (BST) for fast lookup. Insert the following IDs, in the given order, into an initially empty BST: 45, 25, 65, 15, 35, 55, 75, 30, 40.

(i) Draw the final tree (3 Marks).

(ii) Write its pre-order, in-order and post-order traversals (3 Marks).

(iii) What do you observe about the in-order output, and why does it happen? (2 Marks)

(8 Marks)

b) i)Write the recursive algorithm to insert a key into a BST (a private function insert (i, t) and the public function that calls it).

ii)Explain why a new node is always inserted as a leaf, and show the sequence of recursive calls when the key 50 is inserted into the tree drawn in part (a). (6 Marks)

c) Apply (i) Bubble Sort and (ii) Selection Sort on the array [29, 10, 14, 37, 13].

Show the array after every pass for both algorithms. Then state the best-case and worst-case time complexity of each, and which of the two performs fewer swaps. (6 Marks)

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